and privacy mechanisms including a 7-byte unique chip-identifier code, TruST25 digital signature, NFC Forum T2T permanent write locks at block level, and a configurable kill mode that permanently deactivates the tag. Additional benefits include a wide operating-temperature range of -40°C to +85°C and long data retention.

STMicroelectronics

www.st.com

Non-isolated offline switcher IC

The AP3928 is a universal AC high-voltage input non-isolated offline switcher IC that addresses the challenges associated with using a bias

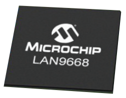
power supply. It delivers safe and reliable power, while simultaneously maintaining elevated performance figures and enabling significant cost savings. This switcher IC is optimised for situations where higher power levels are mandated, such as in industrial control or home automation systems, home appliances and IoT devices. A key differentiator of this high-voltage switcher IC is that it supports conventional offline non-isolated flyback configurations as well as more streamlined non-isolated buck configurations, keeping the bill-of-materials (BOM) to a minimum.

DIODES Incorporated

www.diodes.com

GaN MMIC power amplifier

The new GMICP2731-10 GaN MMIC power amplifier helps deliver high RF output power while simultaneously ensuring the data signals retain their desired characteristics. The GMICP2731-10 is fabricated using GaN-on-silicon carbide (SiC) technology. It delivers up to 10W of saturated RF output power across the 3.5GHz



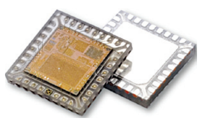
bandwidth between 27.5 and 31GHz. Its power-added efficiency is 20%, with 22dB small-signal gain and 15dB return loss. A balanced architecture allows the GMICP2731-10 to be well matched to 50 ohms and includes integrated DC blocking capacitors at the output to simplify design integration. It is designed for use in commercial and defence satellite communications, 5G networks, and other aerospace and defence systems.

Microchip Technology

www.microchip.com

Optical encoder sensors

The FlexSense series is a revolutionary technology designed to optimise



optical encoder applications and consists of the FS210 transmissive incremental encoder and the FS310 reflective incremental encoder sensors. The FlexSense encoder sensor array vastly reduces the design complexities of optical encoder solutions and integrates several key features including auto-alignment capability, a closed-loop LED driver, on-chip diagnostics, and a state-of-the-art 8x interpolator. In addition, the fully programmable array system on chip (SOC) provides a versatile solution that can adapt to a multiple range of code-disk diameters and pulses per revolution as well as several LEDs.

TT Electronics

www.ttelectronics.com

MODULES & BOARDS

IoT starter kit

The Wiliot Starter Kit contains everything retailers need to experience the IoT transformation. It includes postage stamp sized computers—the

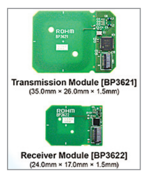
IoT Pixels—pre-mounted on both cardboard and a coffee cup. The IoT Pixels can sense a range of physical and environment data, such as temperature, fill level, motion, location changes, humidity, and proximity. The kit also includes small Bluetooth bridge devices that energise and read the IoT Pixel tags. The harvested data is relayed to the Wiliot Cloud via gateways. Through the starter kit, retailers can explore projects that enable a host of new retail use cases—from enhanced inventory management to consumption monitoring, anti-counterfeiting, product traceability, and more.

Wiliot

www.wiliot.com

Wireless charger modules

The BP3621 (transmitter) and the BP3622 (receiver) modules allow



adding wireless power supply functionality to smaller devices, such as smart tags/cards or PC peripherals. The 13.56MHz wireless charging modules allow users to easily add wireless power functionality to thin and compact devices. Measuring just 20mm² to 30mm² each, the BP3621 and BP3622 incorporate an optimised antenna (coil) layout. It